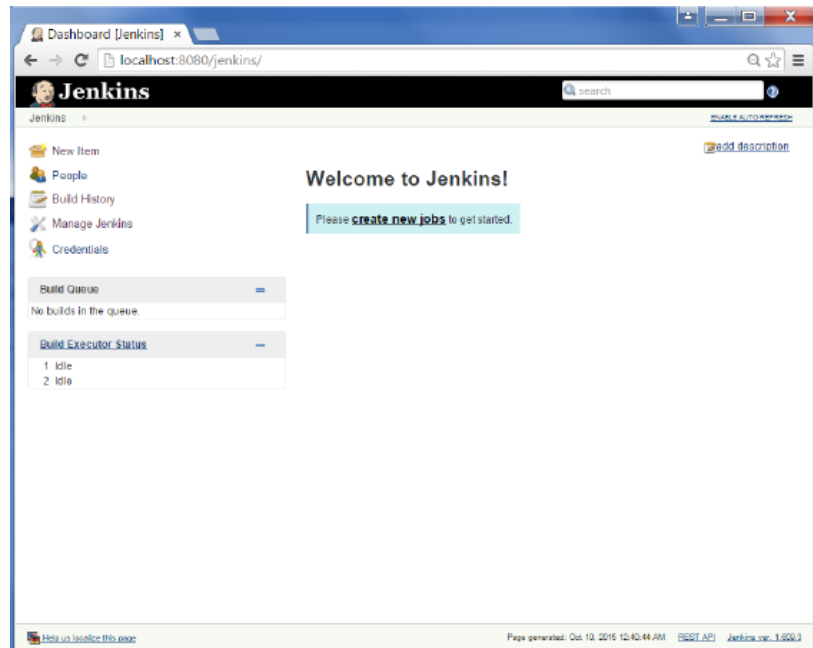




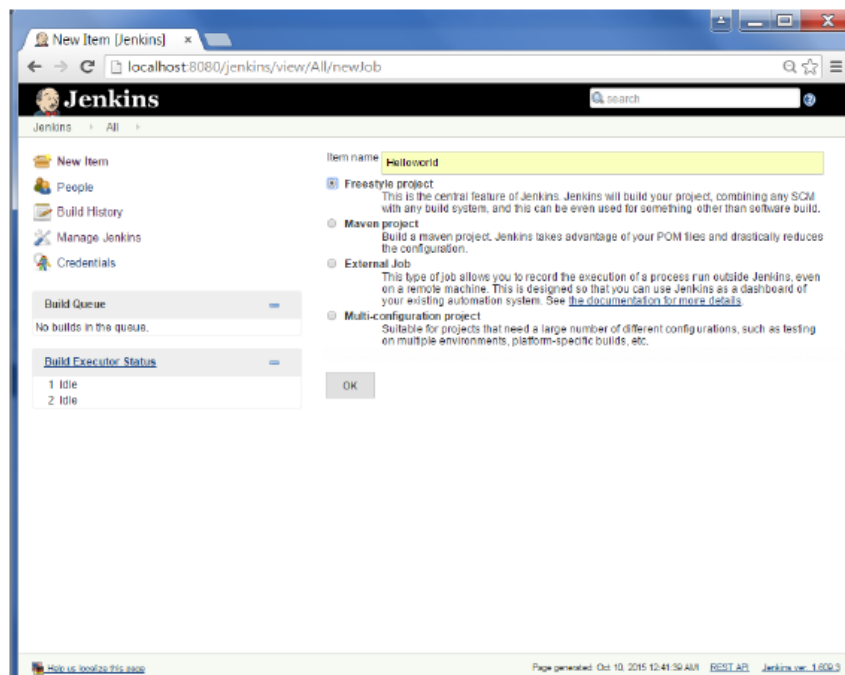
Jenkins

first Job Creation of Jenkins :-

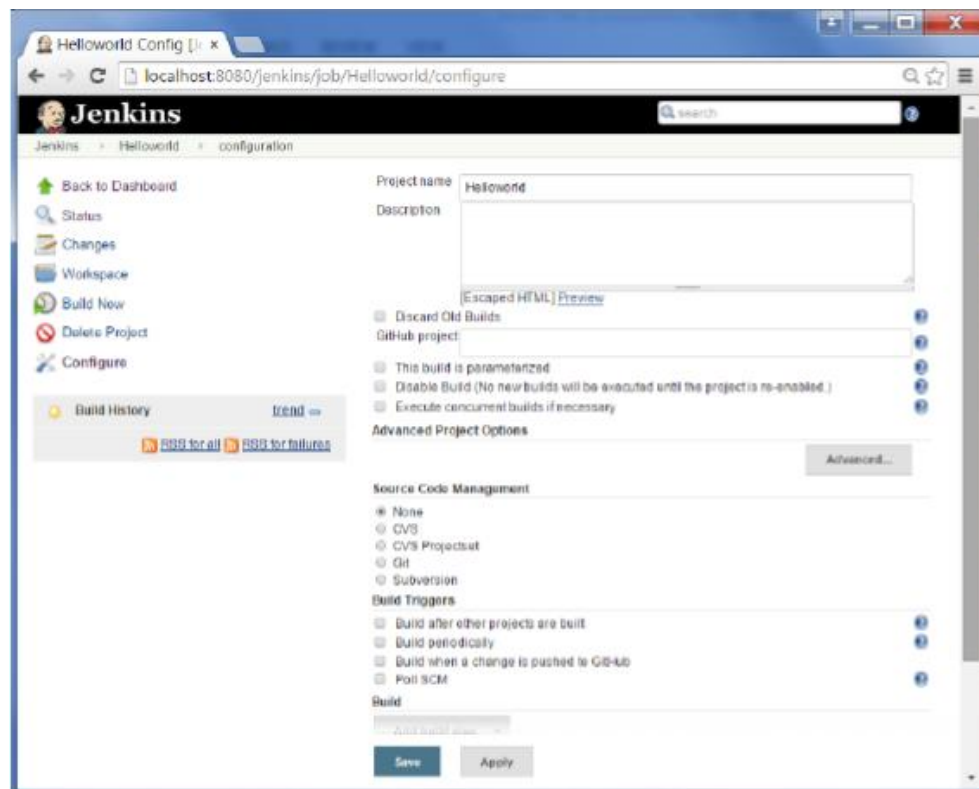
Step 1 – Go to the Jenkins dashboard and Click on New Item



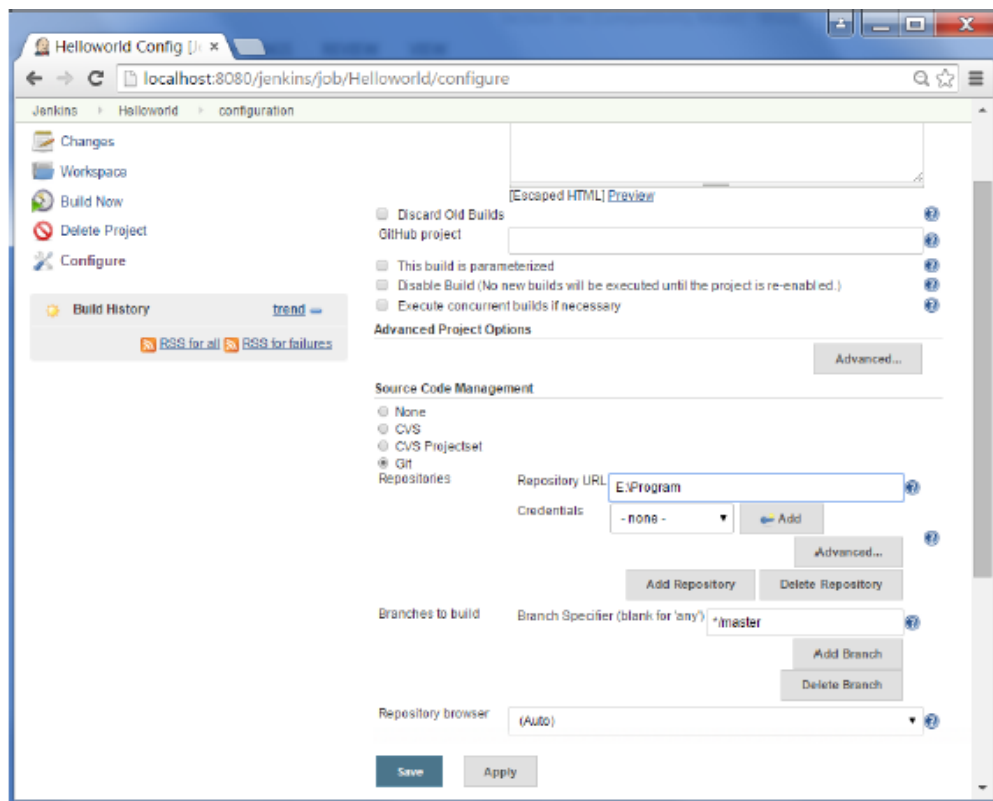
Step 2 – In the next screen, enter the Item name, in this case we have named it Helloworld. Choose the Freestyle project option



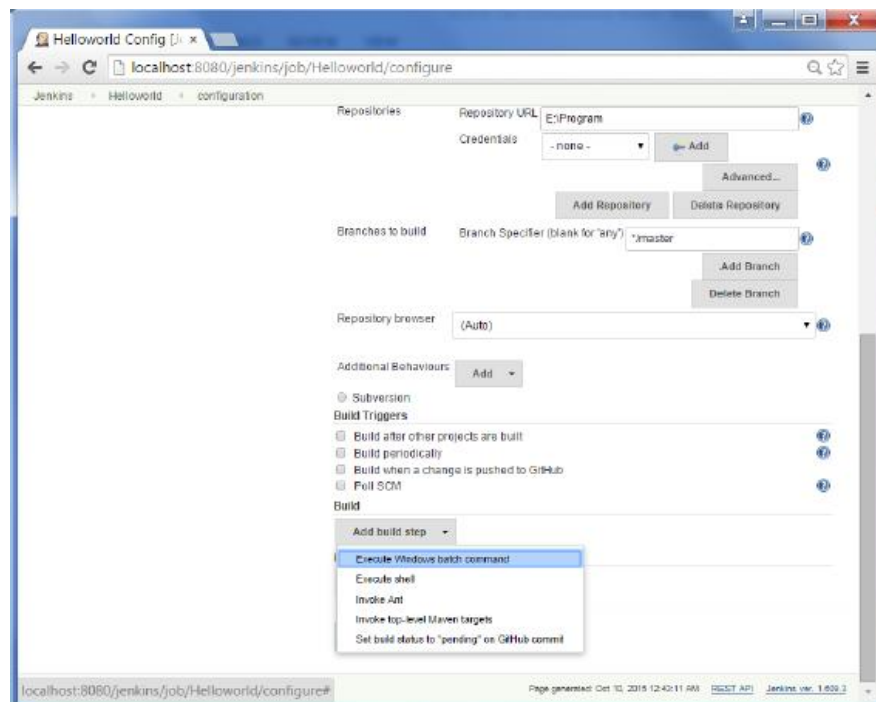
Step 3 – The following screen will come up in which you can specify the details of the job.



Step 4 – Select Git as the source and enter the path to your local repository (e.g., E:\Program) that contains the HelloWorld.java file.



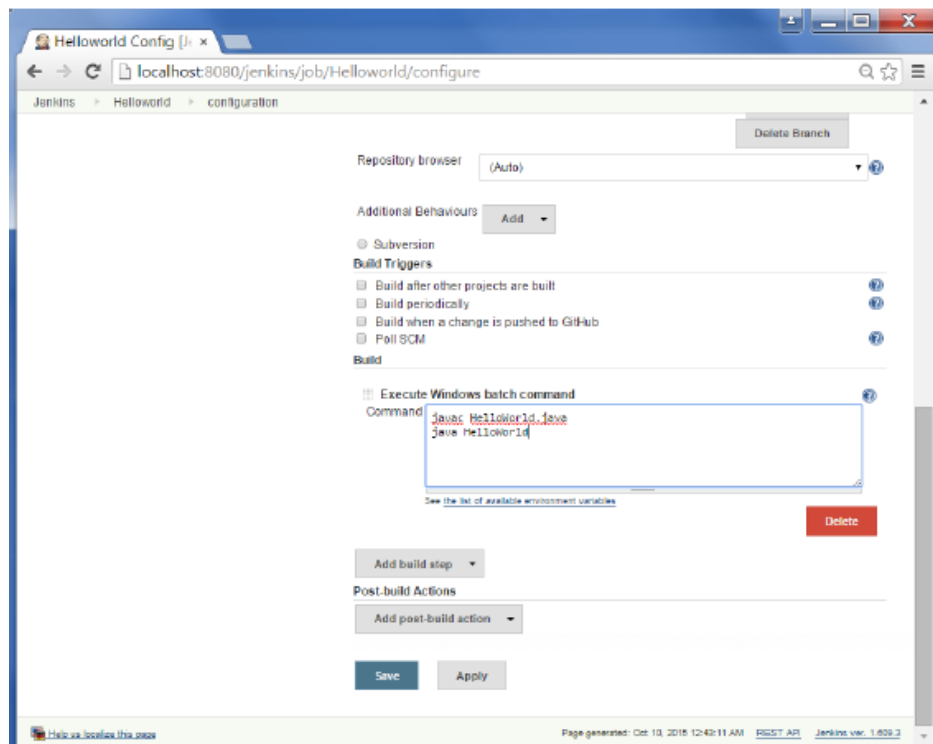
Step 5 – Now go to the Build section and click on Add build step → Execute Windows batch command



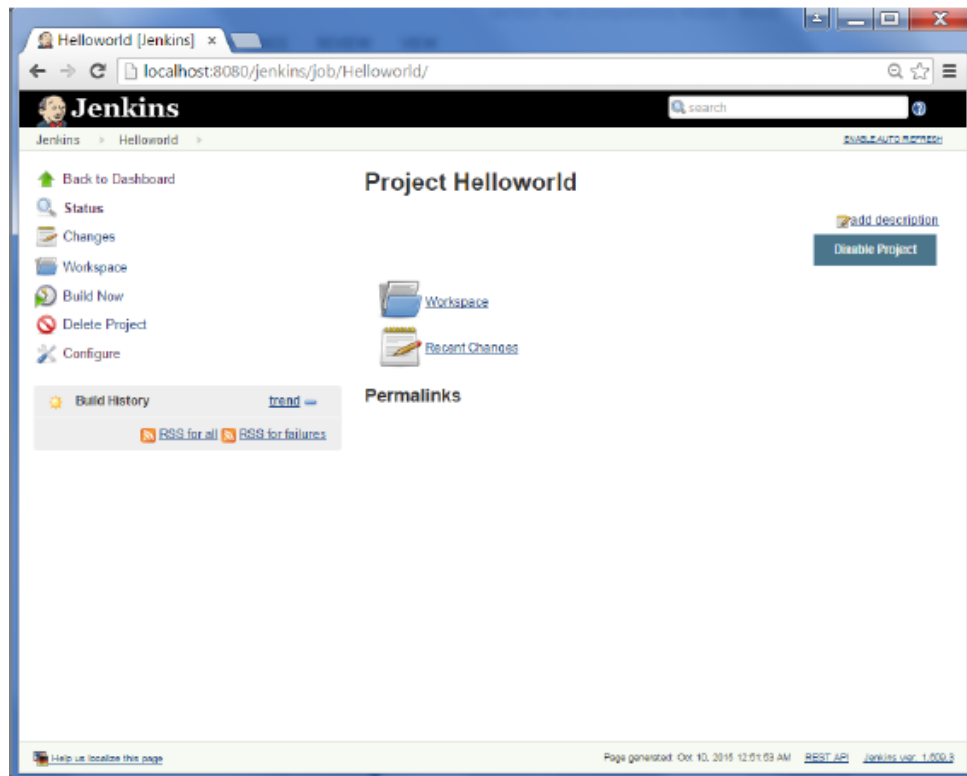
Step 6 – In the command window, enter the following commands and then click on the Save button.

Javac HelloWorld.java

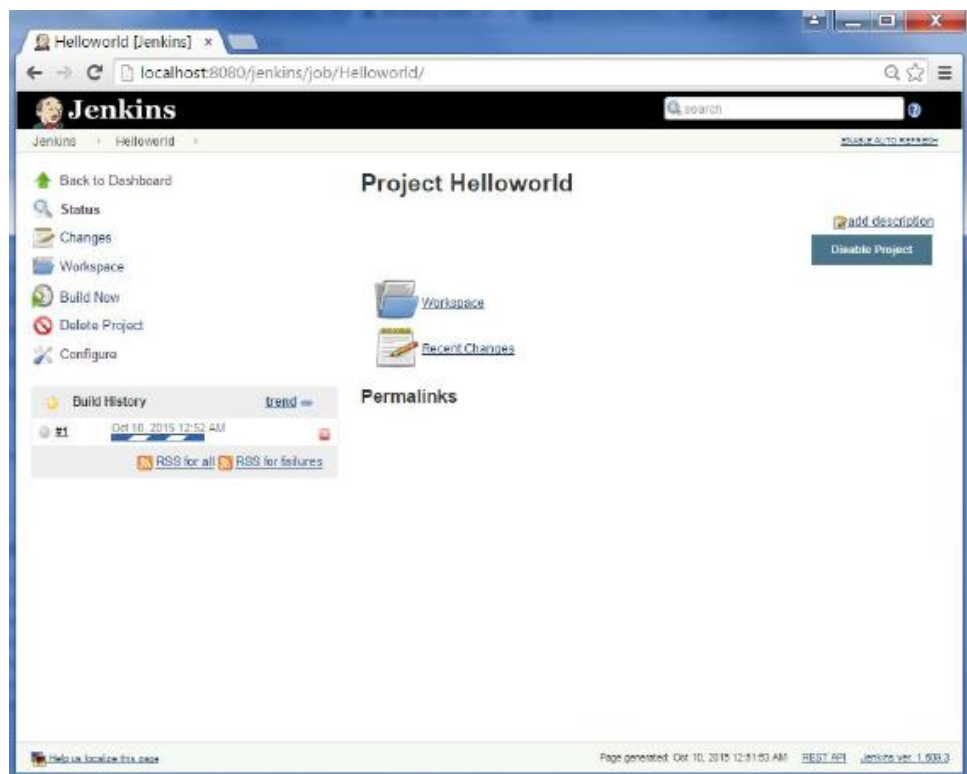
Java HelloWorld



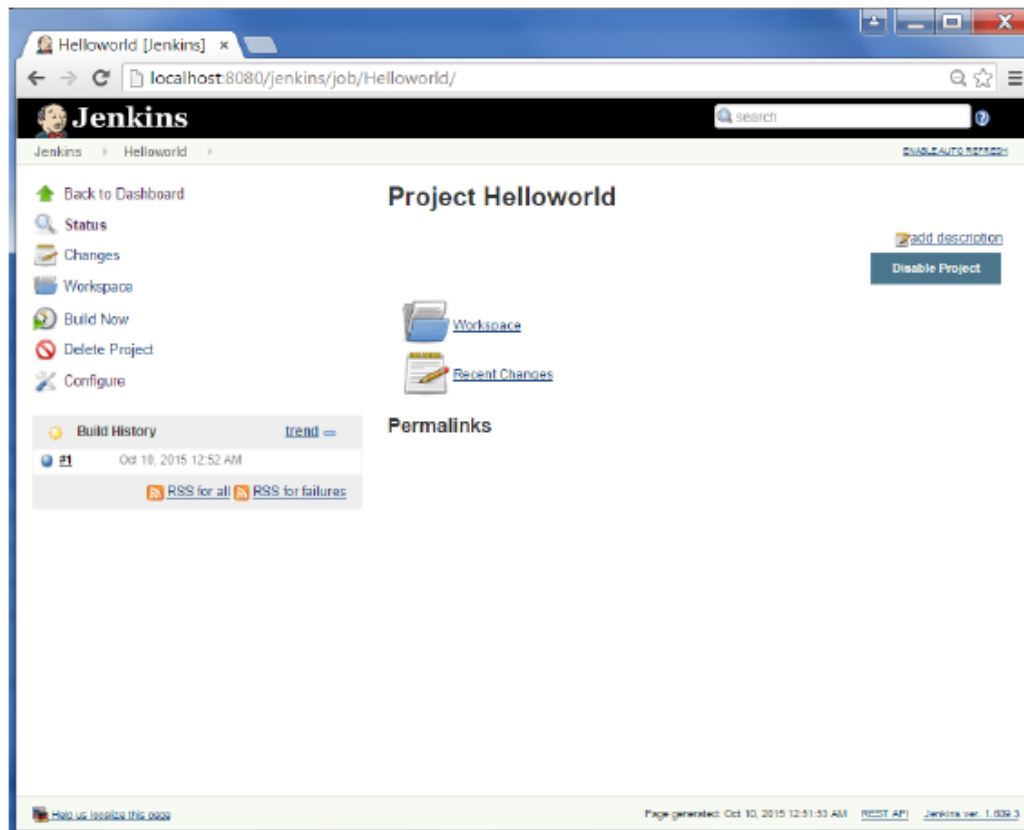
Step 7 – Once saved, you can click on the Build Now option to see if you have successfully defined the job.



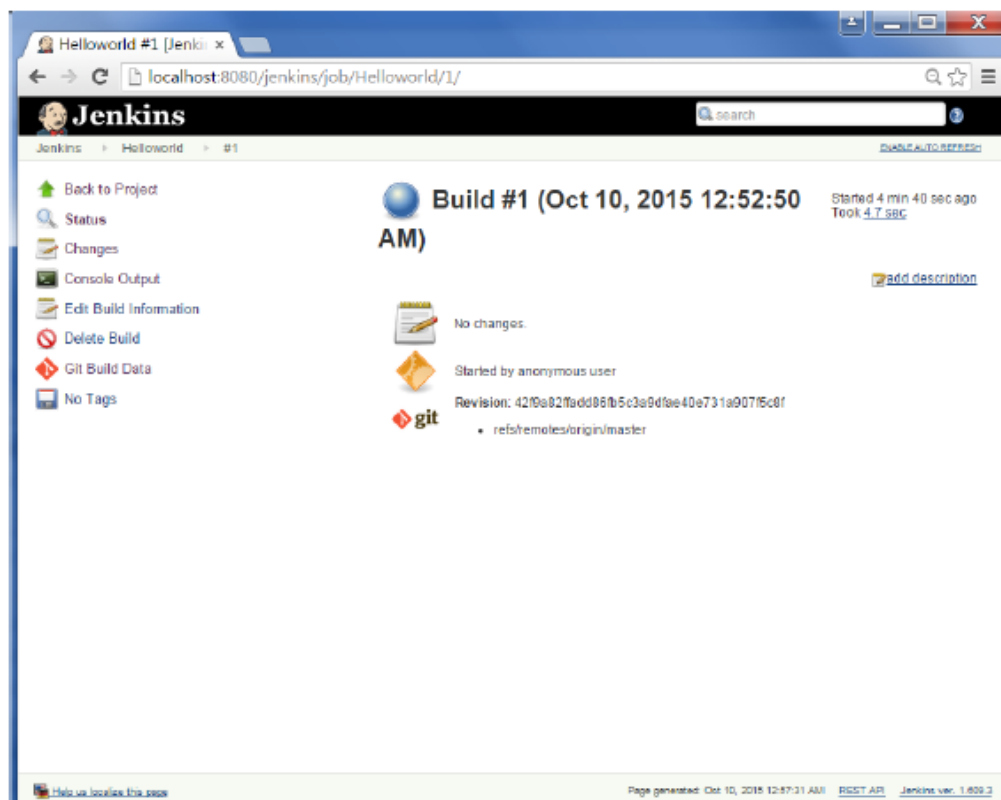
Step 8 – Once the build is scheduled, it will run. The following Build history section shows that a build is in progress.

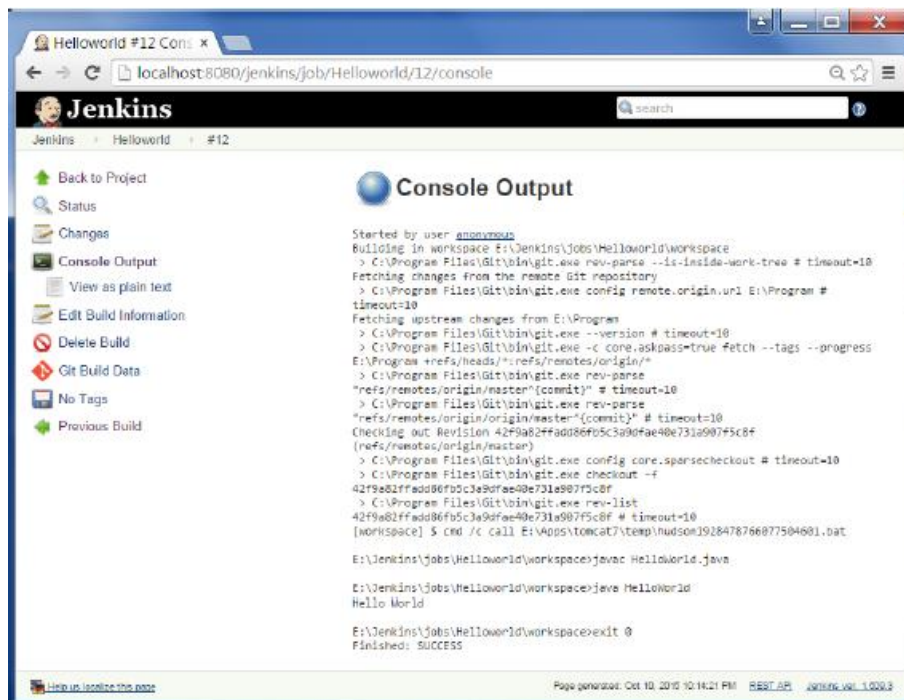


Step 9 – After the build finishes, Jenkins will show whether it succeeded. Click on #1 in the Build History to view the build details.



Step 10 – Click on the Console Output link to see the details of the build





Jenkins provides multiple ways to create build jobs, making it a highly flexible continuous deployment tool.

Building Freestyle Projects In Jenkins :-

Jenkins Freestyle Project allows you to easily create and build jobs without coding, using simple console-based configuration.

Here are the steps to create a freestyle project in Jenkins.

Step 1: Access Jenkins and Log In



Sign in to Jenkins

Invalid username or password

Username

Password

☐ Keep me signed in

Sign in

Step 2: Create a New Freestyle Project

Enter an item name

» Required field



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates separate namespace, so you can have multiple things of the same name as long as they are in different folders.



Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.



Organization Folder

Creates a set of multibranch project subfolders by scanning for repositories.

Step 3: Configure the Project

Dashboard > GFG-Project > Configuration

Configure

- General
- Source Code Management
- Build Triggers
- Build Environment
- Build Steps
- Post-build Actions

General

Description

Plain text [Preview](#)

- ☐ Commit agent's Docker container ?
- ☐ Define a Docker template
- ☐ Discard old builds ?
- ☐ GitHub project
- ☐ This project is parameterized ?
- ☐ Throttle builds ?

[Save](#) [Apply](#)

Step 4: Configure the GitHub Repository

Dashboard > pipeline > Configuration

Advanced ▾

Configure

- General
- Source Code Management
- Build Triggers
- Build Environment
- Build Steps
- Post-build Actions

Source Code Management

☐ None

☒ Git ?

Repositories ?

Repository URL ?

Please enter Git repository.

Credentials ?

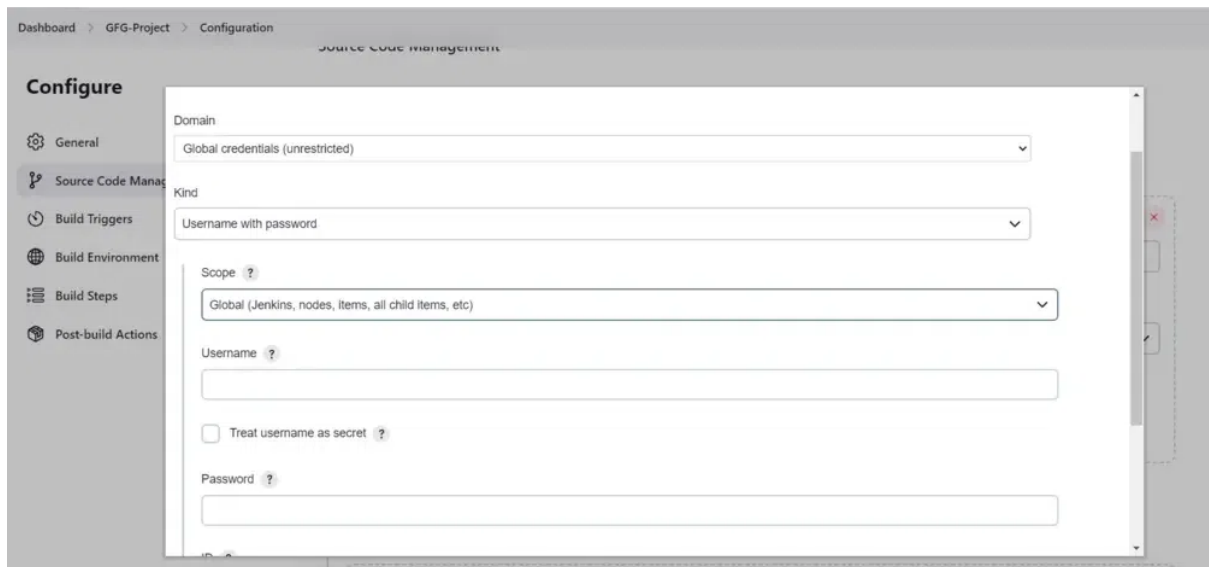
- none - ▾

+ Add ▾

Advanced ▾

[Save](#) [Apply](#)

Step 5: Add Credentials for GitHub



The screenshot shows the Jenkins Configuration page for Source Code Management. The left sidebar contains a menu with options: General, Source Code Management (selected), Build Triggers, Build Environment, Build Steps, and Post-build Actions. The main content area is titled 'Configure' and contains the following fields:

- Domain:** A dropdown menu with the selected value 'Global credentials (unrestricted)'.
- Kind:** A dropdown menu with the selected value 'Username with password'.
- Scope:** A dropdown menu with the selected value 'Global (Jenkins, nodes, items, all child items, etc)'.
- Username:** A text input field.
- Treat username as secret:** A checkbox that is currently unchecked.
- Password:** A text input field.

Step 6: Select Git Branch

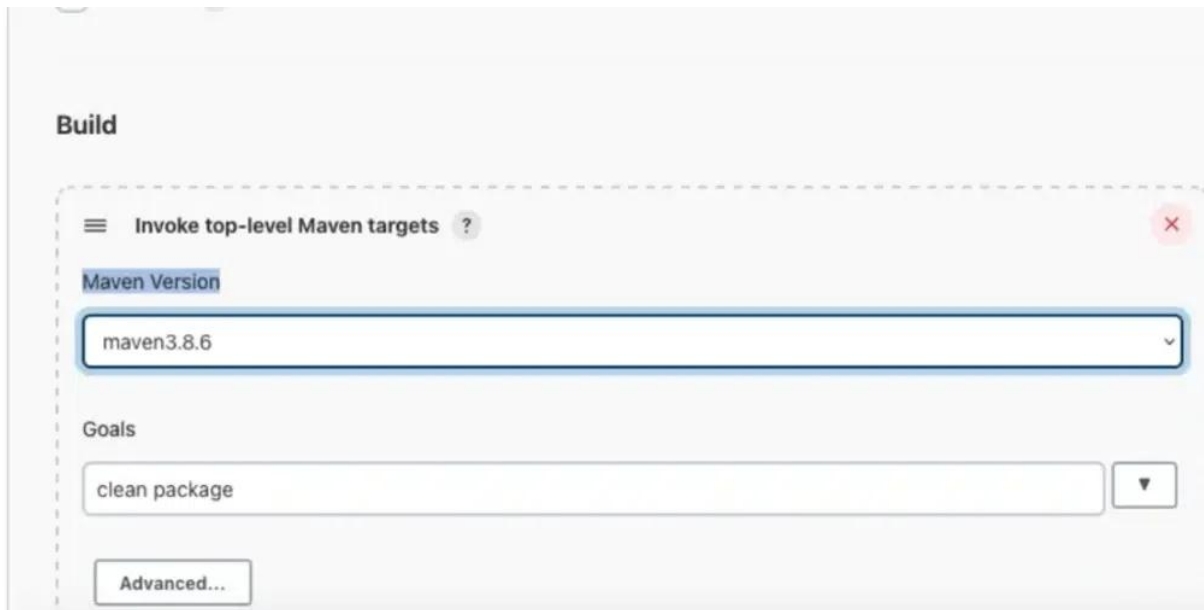
In the Branches to build section, specify the branch you want Jenkins to clone.



The screenshot shows the Jenkins Configuration page for Source Code Management, specifically the 'Branches to build' section. The section is titled 'Branches to build' and contains the following elements:

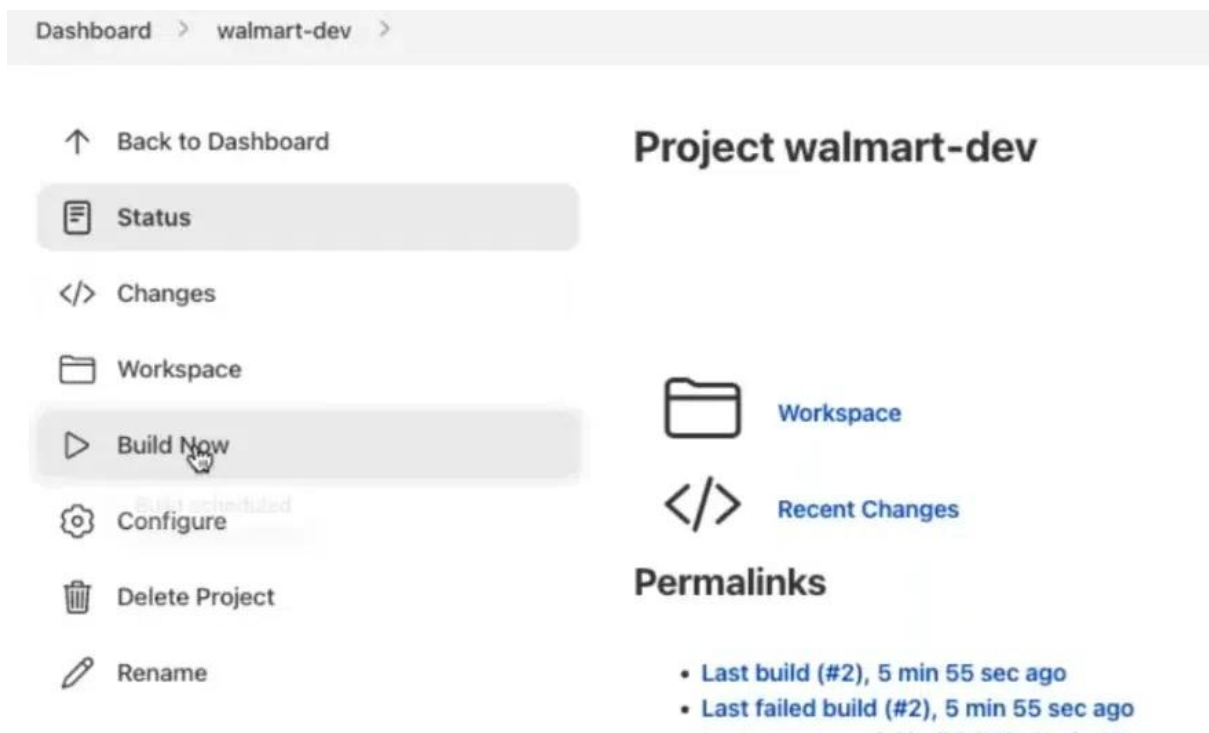
- Add Repository:** A button at the top of the section.
- Branch Specifier (blank for 'any'):** A text input field with the value `*/master`.
- Add Branch:** A button at the bottom of the section.

Step 7: Configure Maven Build



The screenshot shows the 'Build' configuration window for Maven. At the top, it says 'Invoke top-level Maven targets'. Below this, the 'Maven Version' is set to 'maven3.8.6'. Under the 'Goals' section, 'clean package' is entered. There is an 'Advanced...' button at the bottom left of the configuration area.

Step 8: Save the Configuration



The screenshot shows the project dashboard for 'walmart-dev'. On the left, a sidebar contains navigation options: 'Back to Dashboard', 'Status' (selected), 'Changes', 'Workspace', 'Build Now' (highlighted with a mouse cursor), 'Configure', 'Delete Project', and 'Rename'. The main area on the right is titled 'Project walmart-dev' and includes sections for 'Workspace' (with a folder icon), 'Recent Changes' (with a code icon), and 'Permalinks'. The 'Permalinks' section lists two items: 'Last build (#2), 5 min 55 sec ago' and 'Last failed build (#2), 5 min 55 sec ago'.

Step 9: Monitor the Build

The screenshot shows the Jenkins Build Monitor interface. On the left, there are buttons for 'Build Now', 'Configure', 'Delete Project', and 'Rename'. On the right, there are links for 'Workspace' and 'Recent Changes'. Below these is a section titled 'Permalinks' with a list of build links: 'Last build (#2), 5 min 55 sec ago', 'Last failed build (#2), 5 min 55 sec ago', 'Last unsuccessful build (#2), 5 min 55 sec ago', and 'Last completed build (#2), 5 min 55 sec ago'. At the bottom, there is a 'Build History' section with a search bar and a table of builds. The table shows build #3, which is the most recent and is in a failed state (indicated by a red 'x' icon). The build time is 'Jul 7, 2022 2:44 AM'.

Build Now

Configure

Delete Project

Rename

Workspace

Recent Changes

Permalinks

- Last build (#2), 5 min 55 sec ago
- Last failed build (#2), 5 min 55 sec ago
- Last unsuccessful build (#2), 5 min 55 sec ago
- Last completed build (#2), 5 min 55 sec ago

Build History trend

Filter builds...

#3 Jul 7, 2022 2:44 AM

Step 10: Review the Build Status

```
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-filtering/1.0-beta-2/maven-filtering-1.0-beta-2.jar (33 kB at 50 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/xpp3/xpp3_min/1.1.4c/xpp3_min-1.1.4c.jar (25 kB at 38 kB/s)
[INFO] Packaging webapp
[INFO] Assembling webapp [maven-web-application] in [/var/lib/jenkins/workspace/walmart-dev/target/maven-web-application]
[INFO] Processing war project
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/walmart-dev/src/main/webapp]
[INFO] Webapp assembled in [45 msecs]
[INFO] Building war: /var/lib/jenkins/workspace/walmart-dev/target/maven-web-application.war
[INFO] WEB-INF/web.xml already added, skipping
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:23 min
[INFO] Finished at: 2022-07-07T02:45:31Z
[INFO] -----
Finished: SUCCESS
```

We created and configured a Freestyle project in Jenkins, connected it to your GitHub repository, and set up Maven for building your project.